



Enhanced landfill leachate treatment performance by adsorption-assisted membrane distillation

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ABSTRACT

Membrane fouling and high-strength membrane concentrate production are two limitations of membrane distillation (MD) for landfill leachate treatment. In this study, activated carbon- and biochar-based adsorption processes were integrated into a conventional MD system to overcome these limitations. The organic matter fractionations of the leachate were thoroughly investigated during the treatment. Membrane-reversible and irreversible foulants differed remarkably from the inlet leachate in the non-assisted MD system. Specifically, reversible foulants were characterized by a high abundance of humic-like fluorescent components, high-molecular-weight humic-size constituents, peptides, and unsaturated compounds. In contrast, irreversible foulants were enriched with fulvic-like fluorescent components, low-molecular-weight neutrals, unsaturated compounds, and polyphenols. The adsorption-based pre-treatment effectively removed foulant precursors from landfill leachate, with a relatively higher (20%) adsorption performance for specific biochar used in this study than for activated carbon. Compared with the non-assisted MD system, the biochar-assisted MD system showed improved performance, achieving 40% overall membrane flux recovery, 42% higher filtration fluxes, and 53% lower concentrate production. In addition, a 15% higher removal of irreversible foulants was observed as compared to the reversible foulants, which can potentially increase the membrane lifespan. This study demonstrates the effectiveness of an adsorption-assisted MD system supported by increased filtration, membrane fouling alleviation, and low-strength leachate concentrate generation.

1. Introduction

Landfill leachate is a heterogeneous mixture of complex organics and inorganics. As it poses a direct hazard to the environment (Jovanov et al., 2018; Kjeldsen et al., 2002), landfill leachate requires prompt and careful measures before its disposal (Liu et al., 2015). Membrane separation processes are considered a smart solution for the onsite treatment of leachate and disposal of fine-quality effluent (Mariam and Nghiem, 2010; Silva et al., 2018; Trebouet et al., 2001). However, high operational costs and severe membrane fouling are two downsides of this technology (Her et al., 2007; Zularisam et al., 2006). Recently, membrane distillation (MD) has been shown to be a cost-effective treatment option, because MD systems utilize heat energy instead of the high trans-membrane pressure that is required for conventional membrane

treatments, such as reverse osmosis (RO) and nanofiltration (NF) (Yan et al., 2021). MD has been effectively applied to treat many complex high-strength wastewaters, including landfill leachate (Asif et al., 2021b; Tijjing et al., 2015; Zico et al., 2021). The energy requirements for the MD are 12 times lower than the reverse osmosis process (López-Porfiri et al., 2023) and the treatment efficiency for MD for landfill leachate is reported to be 12% higher than the RO (Chaudhari and Murthy, 2010; de Souza Marotta Alfaia et al., 2023). However, subsequent organic membrane fouling has been extensively reported and is a drawback of MD systems in treating various landfill leachates (Alkhatib et al., 2021).

To address the limitations of organic membrane fouling, several integration options (e.g., advanced oxidation, coagulation-flocculation, ion exchange resins, and adsorption processes) with conventional

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